

SEDAR 50 Atlantic Blueline Tilefish

Stock ID

Stock ID Work Group

Genetics

- Given the results of the genetic work on Blueline Tilefish evaluated here and the limitations identified in the Katz et al. 1983 (SEDAR50-RD18) Golden Tilefish study, patterns in genetic population structure should be revisited for other deep-water species (including Golden Tilefish) using contemporary genetic approaches and analyses.
- To develop a mechanistic understanding of processes facilitating gene flow for Blueline Tilefish, further research should be undertaken to evaluate spawning season duration, pelagic larval stage duration, and adult movements.
- Additional genetic sampling should be conducted in the Gulf of Mexico (Florida Keys to the Texas-Mexico border) to further evaluate the potential for genetic structure across the Gulf of Mexico.

Life History

- Age reading interpretation of Blueline Tilefish otoliths need to be resolved. Other age validation techniques should be investigated (e.g., Pb/Ra ratio).
- Reproductive biology studies of Blueline Tilefish should be expanded to include the full distributional range of the species, specifically targeting samples from the west and east coasts of Florida and the Mid-Atlantic region. These data are needed to assess possible shifts in spawning season. Sampling of young fish is needed to improve the maturity ogive.
- Better information is needed on the movement or migration of juvenile and adult Blueline Tilefish.
- Studies should be conducted on the identification of Blueline Tilefish larvae and also on the location, duration, and dispersal mechanisms of the egg and larval stages.

Spatial Distribution

- Further research should be conducted to understand the thermal tolerance of Blueline Tilefish.
- Surveys should be conducted to try to document the distribution of early life stages.
- Further studies are needed on habitat preferences over the whole range of the species.
- Particle modeling to investigate hypotheses about movement of eggs and larvae.
- Research into movement of adults.

Overall

- A continuous, random, stratified survey should be developed and implemented for Blueline Tilefish throughout its range.

Joint SSC Sub-Panel

- The Panel recognizes the limited information to fully describe the population structure of Blueline Tilefish throughout the species' range in U.S. waters, and therefore endorses the research recommendations from the Stock ID work group report.

Science/Management Leadership Group

- The group also recommends that research be conducted to fully define population structure of Blueline Tilefish in the Gulf of Mexico, and to define the relationship between fish in the Gulf of Mexico with those in the Atlantic.

Data Workshop

Life History Working Group

- Collect and take reproductive tissue samples from smaller fish to improve reproductive parameters estimates.
- Investigate movements and locations for post-settlement smaller/juvenile Blueline Tilefish.
- Investigate adult movement through tagging studies (e.g., breakaway tags; see SEDAR50_DW12)
- Design and implement a regional ichthyoplankton survey to investigate larval transport. Note: taxonomic work needs to be done first to describe the eggs and larvae of Blueline Tilefish.
- Mine existing ichthyoplankton collections be assessed for presence of Blueline Tilefish larvae.
- Collect information/data on reproductive and larval behavior for use in modelling larval dispersal.
- Studies to validate the annulus formation and annulus structure of in blueline.
- Further investigate the potential shift in the Radio Bomb Carbon data and reference curve for Blueline Tilefish age validation (Note that this work is ongoing at SCDNR).
- Develop and recommend use of standardized aging methods as recommended by the SEDAR Best Practices Standing Panel Language in the Data Issue Inventory: **Age determination:** develop best practices for age determination to include processing and reading age structures, age calibration, age variability and bias estimates, validation methods, etc.

- Develop and recommend use of methods to provide growth parameter and natural mortality estimates in cases where no reliable age data are available. Focus should be on acceptable approaches for parameter values and error distributions (e.g. meta analyses, use of related species, use of species with comparable life history strategies, etc.).

Discard Mortality Ad-hoc Group

- The working group identified limited peer-reviewed literature for deepwater reef fish species and no information for Blueline Tilefish in either the commercial and recreational fisheries. Future research should attempt to provide estimates of discard mortality through tag-recapture, acoustic tagging, or other methods in both sectors. While some information was available to estimate immediate mortality, research is needed to reduced uncertainty in estimates of delayed mortality. Particular interest was expressed into developing mortality estimates when using descender devices to aid recompression, since these devices may have the potential to substantially lower mortality rates. Cooperative research with either sector represents a robust mechanism available to begin obtaining more estimates of mortality and reduce uncertainty in future assessments.

Commercial Working Group

- Investigate improvements in proportioning unclassified tilefish to species
 - Investigate alternative methods of determining proportions, e.g. relationship to landings of non-tilefish species such as Snowy Grouper
- Increase observer coverage in the South Atlantic
 - Observer data would improve discard estimation and provide estimates of discard sizes and weights
- Implement electronic monitoring of bycatch
 - Such a program should improve discard estimation accuracy and provide size and weight composition of discards

Recreational Working Group

- Research and implement rare-event data collection procedures.
 - E.g. mandatory reporting, logbooks, reef fish stamp to determine universe.
- Fund research efforts to collect discard length and age data from the private sector.
- Additional data collection in the recreational fishery (gear, depth, angler demographics)
- Pre-stratify MRIP Keys, N-S Canaveral, N-S Hatteras.
- At-sea observers collect surface and bottom temperature.

Index Working Group

- The IWG discussed future research recommendations for Blueline Tilefish. The unanimous consensus was that a coastwide fishery-independent survey is needed for Blueline Tilefish. In the absence of a fishery-independent index, additional information

on the targeting behavior of fishermen, in particular the depth or geographic locations fished within a given trip as well as more refined information on fishing effort is needed.

ToR#7 Ad-hoc Group

ToR #7 Consider ecosystem and climate issues that could affect population dynamics. Identify and describe available data sources to investigate the effects of abiotic and biotic factors, for example climate change, predator/prey interactions, etc., on recruitment, growth, geographic distribution, and natural mortality.

Initiate studies to:

- describe movements/migration of adult Blueline Tilefish,
- investigate possibility of range expansion using recent statistical models and available data,
- determine thermal tolerance of Blueline Tilefish,
- identify Blueline Tilefish larvae,
- investigate larval duration and larval dispersal,
- identify juvenile habitat or movement,
- collect temperature within the water column,
- collect information on location of life stage activities.

Assessment Workshop

- Reliable fishery independent indices should be developed, though this would first require a fishery independent survey that samples Blueline Tilefish effectively.
- Aging techniques should continue to be developed for Blueline Tilefish so that future stock assessments may be done using age structured models.
- Genetics samples should be collected from the West Florida Shelf and other areas throughout the Gulf of Mexico to more convincingly determine whether or not Blueline Tilefish in the Gulf of Mexico are part of the same population as Atlantic Blueline Tilefish.
- The search for small blueline tilefish ($< 25cm$), as well as eggs and larvae, should continue. Blueline Tilefish $< 25cm$ are bound to be much more abundant than larger individuals and yet they are rarely encountered by any gear. Eggs and larvae can effectively be identified using genetic techniques, and there are apparently many samples that have been collected but have not yet been genotyped (Lewis et al. 2016). A good place to start would be genotyping more of the available samples. This would also benefit the science on other species.

- Any information on movement of adult Blueline Tilefish, especially movement across Council boundaries would be valuable (e.g. tagging studies). It has been shown that other deepwater tilefish species can be tagged and recaptured (Grimes et al. 1983).
- The possible movement of Blueline Tilefish eggs and larvae between Council regions via ocean currents should be investigated further, perhaps with particle tracking models, or more in depth drifter analysis.

Review Workshop

- *Clearly denote research and monitoring that could improve the reliability of, and information provided by, future assessments.*

Research recommendations were provided by the Data and Assessments workshops and were reviewed at the Review Workshop. The two main areas where further research would help improve the assessment for both stocks are development of fishery independent indices and resolution of the age determination issues. A third area of research pertains to improving our understanding of biological processes such as reproduction and recruitment dynamics.

NMFS should continue the development of fishery dependent and independent indices for these stocks. The development of a continuous, random, stratified fishery-independent survey implemented for blueline tilefish throughout its range is potentially the single most important recommendation for this stock. The survey could also be used to fill other knowledge gaps (e.g. incorporate a hook selectivity study).

Reliable age reading or growth curve development is also urgently needed for blueline tilefish. If possible, the age reading issues of blueline tilefish otoliths should to be resolved. Other age validation techniques or methods to derive reasonable growth curves should be investigated (e.g., Pb/Ra ratio, or tagging studies). Resolution of the age determination issues would allow considerable past information to be incorporated into the assessment.

Beyond those priorities, further understanding of egg and larval dispersal through biophysical modeling and genetic analysis may be useful. Studies should be conducted on the identification of blueline tilefish larvae and also on the location, duration, and dispersal mechanisms of the egg and larval stages. Eggs and larvae can effectively be identified using genetic techniques, and there are apparently many samples that have been collected but have not yet been genotyped. A good place to start would be genotyping more of the available samples. This would also benefit the science on other species.

Increased observer coverage for both commercial and recreational fisheries in the South Atlantic would improve discard estimation and provide estimates of discard sizes and weights. Implementing electronic monitoring of bycatch would improve discard estimation accuracy and

provide size and weight composition of discards. In the absence of a fishery-independent index, additional information on the targeting behavior of fishermen, in particular the depth or geographic locations fished within a given trip as well as more refined information on fishing effort is needed.

Investigation of alternative methods in proportioning unclassified tilefish to golden, blueline, or other species could be explored. For recreational fisheries, research into and implement rare-event data collection procedures (e.g. mandatory reporting, logbooks, reef fish stamp to determine universe.) could also benefit the assessment. ,

An increase in sample size on the catch compositions from both commercial and recreational fisheries would be useful.

Estimates of immediate and delayed discard mortality through tag-recapture, acoustic tagging, or other methods in both commercial and recreational fisheries are needed. Special interest was expressed into developing mortality estimates when using descender devices to aid recompression, since these devices may have the potential to substantially lower mortality rates

Special effort could be put into the data collection in North and South of Cape Hatteras to better understand the spatial distribution of fishing effort, size composition of catch, and catch rate.

Further study on the maturity and reproductive biology would be of value to the assessment. Reproductive biology studies of blueline tilefish should be expanded to include the full distributional range of the species, specifically targeting samples from the west and east coasts of Florida and the Mid-Atlantic region. These data are needed to assess possible shifts in spawning season. Sampling of young fish is needed to improve the maturity ogive to improve reproductive parameters estimates.

Particularly if the age determination issues remain unresolved, development and use of methods to provide growth parameter and natural mortality estimates (may serve as prior elicitation in the future) would benefit the assessment. The focus should be on acceptable approaches for parameter values and error distributions (e.g. meta analyses, use of related species, use of species with comparable life history strategies, etc.)

With respect to assessment models, consideration should be given to developing Bayesian, age-structured models to integrate prior knowledge from the meta-analysis on the life history processes.

- *Provide recommendations on possible ways to improve the SEDAR process.*

The current SEDAR process looks well designed. The process provides for a thorough review and evaluation of the available data, provides thorough consideration and review of analytical approaches and modeling results, provides very good guidance on the information expected to result from the process, and provides very good documentation of the process including decisions made throughout the assessment. The process is highly transparent, particularly because documents produced for review remain unedited after the review. The pre-Review Workshop teleconference is a good component that can help get the Review Workshop meeting to quick start by providing the analytical team advance notice of areas that the RW is likely to question. Distributing presentations in advance of the workshop is useful. As a minor recommendation would be to ensure there is time for at least two rounds of review of the Review Workshop report in the event that there are significant additions to the report or addendum material provided to RW after the meeting is adjourned. This would help to ensure that the RW has the opportunity to reach consensus on all aspects of the report.